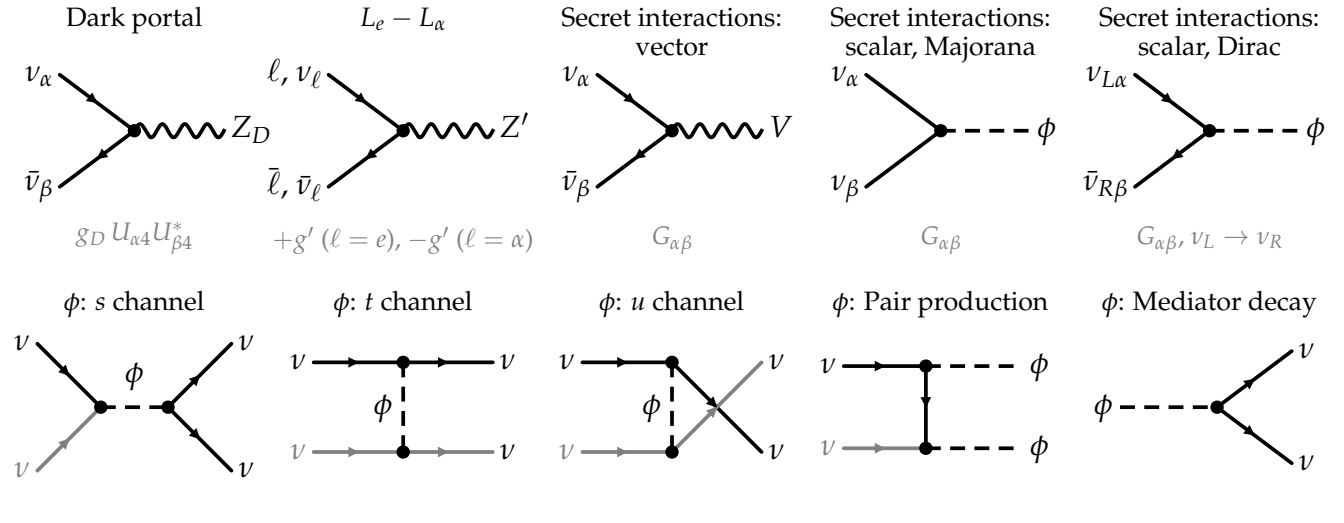

Dark portal
$L_e - L_\alpha$
Secret interactions: vector
Secret interactions: scalar, Majorana
Secret interactions: scalar, Dirac
ν_α
$\bar{\nu}_\beta$
Z_D
ℓ, ν_ℓ
$\bar{\ell}, \bar{\nu}_\ell$
Z'
V
ϕ
$\nu_{L\alpha}$
$\bar{\nu}_{R\beta}$
$g_D\, U_{\alpha 4} U^*_{\beta 4}$
$+g'$ $(\ell = e)$, $-g'$ $(\ell = \alpha)$
$G_{\alpha\beta}$
$G_{\alpha\beta}$
$G_{\alpha\beta}$, $\nu_L \to \nu_R$
ϕ: s channel
ϕ: t channel
ϕ: u channel
ϕ: Pair production
ϕ: Mediator decay
ν
ϕ